Purdue University **2026-2027 University Catalog**

Industrial Engineering, BSIE

Location(s): West Lafayette, Indianapolis

About the Program

The BS program in Industrial Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Industrial Engineering and Similarly Named Engineering Programs.

Industrial engineers design, analyze, and manage complex human-integrated systems such as manufacturing systems, supply chain networks, and service systems. These systems typically consist of a combination of people, information, material, and equipment. In such systems industrial engineers determine how to optimize the system for maximum efficiency, effectiveness, throughput, safety, or some other objective of interest to the stakeholders of the system. To achieve these objectives, an industrial engineer draws upon knowledge of mathematics, along with physical, engineering, management, and behavioral sciences to function as a problem-solver, innovator, designer, coordinator, and system integrator. Industrial engineers are employed in and apply their skills in an extremely wide range of organizations, including manufacturing industries, service industries, and governmental agencies.

The complexity of these organizations and the emphasis on increased effectiveness, efficiency, and productivity have led to a growing need for industrial engineering analysis and design, resulting in an increased demand for industrial engineering graduates. This increased demand recognizes the modern industrial engineer's versatility and responsiveness to the challenges of a rapidly changing society. Industrial engineering is one of the nation's largest and most rapidly growing engineering professions.

The industrial engineering program prepares graduates for careers in all phases of industrial engineering and enables them to perform both technical and managerial functions that require scientific and engineering backgrounds. By combining the study of science, mathematics, engineering fundamentals, design, and management principles, an industrial engineering education provides a unique background and a sound basis for lifelong career development in engineering practice, research, or management.

Senior design projects consist of a real-world application of IE principles by teaming students with a local industry in Indiana. Teams have taken on full-scale projects like designing floor layouts for factories and hospitals, designing operations to improve system efficiency, reducing time and waste in processing, allocating resources to optimize system performance, and developing a safety plan for preventing work-related injuries.

Industrial Engineering Major Change (CODO) Requirements

Degree Requirements

123 Credits Required

Industrial Engineering Major Requirements (51 credits)

Industrial Engineering Required Major Courses (36 credits)

- IE 20000 - Industrial Engineering Seminar **Credit Hours: 0.00**
- IE 23000 - Probability And Statistics In Engineering I **Credit Hours: 3.00**
- IE 33000 - Probability And Statistics In Engineering II **Credit Hours: 3.00**
- IE 33200 - Computing In Industrial Engineering **Credit Hours: 3.00**
- IE 33500 - Operations Research - Optimization **Credit Hours: 3.00**
- IE 33600 - Operations Research - Stochastic Models **Credit Hours: 3.00**
- IE 34300 - Engineering Economics **Credit Hours: 3.00**
- IE 37000 - Manufacturing Processes I **Credit Hours: 3.00**
- IE 38300 - Integrated Production Systems I **Credit Hours: 3.00**
- IE 38600 - Work Analysis And Design I **Credit Hours: 3.00**
- IE 43100 - Industrial Engineering Design **Credit Hours: 3.00**
- IE 47400 - Industrial Control Systems **Credit Hours: 3.00**
- IE 48600 - Work Analysis And Design II **Credit Hours: 3.00**

Industrial Engineering Selectives (15 credits)

Choose one option.

Option I

- IE 47000 - Manufacturing Processes II **Credit Hours: 3.00**
- IE 48400 - Integrated Production Systems II **Credit Hours: 3.00**
- Technical Selective - Credit Hours: 9.00

Option II

- IE 47000 - Manufacturing Processes II **Credit Hours: 3.00** or
- IE 48400 - Integrated Production Systems II **Credit Hours: 3.00**

- Industrial Engineering Selective - Credit Hours: 3.00
- Technical Selective - Credit Hours: 9.00

Please see Industrial Engineering Selectives & Technical Selectives for course options.

Other Program Requirements (72-75 credits)

Other Course Requirements (48-50 credits)

Intro to Engineering I & II - Credit Hours: 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

OR

- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

OR

- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and

- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - **2.00** **Credit Hours**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

AND

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

Physics I - Credit Hours: 0.00-4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00** **Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00** **Credit Hours**

- CS 15900 - C Programming **Credit Hours: 3.00**
- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00**
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00** (C- or better for pre-requisite)
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- ME 20000 - Thermodynamics I **Credit Hours: 3.00**
- ME 27000 - Basic Mechanics I **Credit Hours: 3.00**
- NUCL 27300 - Mechanics Of Materials **Credit Hours: 3.00**
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00**

Engineering General Education Requirements (24-25 credits)

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00

- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements

- *At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)*
- *At least 12 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.*
- *A course is not eligible for use towards the 24 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.*

Supplemental Information

Click here for Industrial Engineering Selectives & Technical Selectives

Click here for Industrial Engineering General Education Selectives

GPA Requirements

- A GPA of 2.0 is required for all Baccalaureate degrees.

Course Requirements and Notes

- No course offered by the Purdue Polytechnic Institute can be counted towards the IE plan of study unless it is used to satisfy a University Core Curriculum foundational learning outcome or otherwise approved as part of the Engineering General Education program.
- Credits taken to meet University Core Curriculum foundational learning outcomes can also be applied toward one area of the BSIE degree requirements (Required Major Courses, Technical Selective, General Education).
- Students are welcome to appeal to have courses reviewed to be added to the approved Technical Selective and/or General Education lists. For additional information about this process, contact an IE academic advisor.

Pass/No Pass Policy

- Courses with a Pass/No Pass grade (P/NP) may not be used to satisfy IE degree requirements.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.

* If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or

- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- CS 15900 - C Programming **Credit Hours: 3.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

16-20 Credits

Fall 2nd Year

- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**
- ME 27000 - Basic Mechanics I **Credit Hours: 3.00**
- IE 20000 - Industrial Engineering Seminar **Credit Hours: 0.00**
- IE 23000 - Probability And Statistics In Engineering I **Credit Hours: 3.00**
- IE 34300 - Engineering Economics **Credit Hours: 3.00**
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

16 Credits

Spring 2nd Year

- IE 33000 - Probability And Statistics In Engineering II **Credit Hours: 3.00**
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- ME 20000 - Thermodynamics I **Credit Hours: 3.00**
- NUCL 27300 - Mechanics Of Materials **Credit Hours: 3.00**
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00**
- General Education II (UCC: BSS) - Credit Hours: 3.00

18 Credits

Fall 3rd Year

- IE 33200 - Computing In Industrial Engineering **Credit Hours: 3.00**
- IE 33500 - Operations Research - Optimization **Credit Hours: 3.00**
- IE 33600 - Operations Research - Stochastic Models **Credit Hours: 3.00**
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- General Education III - Credit Hours: 3.00

15 Credits

Spring 3rd Year

- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00**
- IE 37000 - Manufacturing Processes I **Credit Hours: 3.00**
- IE 38300 - Integrated Production Systems I **Credit Hours: 3.00**
- IE 38600 - Work Analysis And Design I **Credit Hours: 3.00**
- General Education IV - Credit Hours: 3.00

15 Credits

Fall 4th Year

- IE 47400 - Industrial Control Systems **Credit Hours: 3.00**
- IE 48600 - Work Analysis And Design II **Credit Hours: 3.00**
- Technical Selective I - Credit Hours: 3.00
- Technical Selective II - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00

15 Credits

Spring 4th Year

- IE 43100 - Industrial Engineering Design **Credit Hours: 3.00**
- Technical Selective III - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Industrial Engineering Selective - see advisor

15 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are

expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.